Numerical Analysis I

Quiz 6B Name:_____

Use the Taylor expansion to find the linear approximation $L(x)$ to $F(x)$ near x_0, where

$$F : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad \begin{pmatrix} u \\ v \\ w \end{pmatrix} \longmapsto \begin{pmatrix} u^2 + \sin(v) \\ uv + \cos(w) \\ w^2 + u \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 0 \\ \frac{\pi}{2} \\ 0 \end{pmatrix}.$$